

FlyRank AI Internship — Week 1 Deliverable

Assignment: Multi-AI Workflow & Toolstack Handoff Map

Track:	General AI Fluency
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Status:	Ready for Submission

Executive Summary

This deliverable defines the Multi-AI Operational Workflow & Handoff Map for Week 2 of the FlyRank AI Internship: Tool Matrix, Central Hub Architecture in Antigravity, Trigger-Based Handoff Rules, and Socratic AI Tutor Audit.

1. Tool Matrix & Specialization

Tool	Core Specialization	Primary Use Case
Perplexity AI	Live Web Search & Tech Specs	Searching latest framework docs, library updates, and API benchmarks.
Claude	Deep Analytical Reasoning	Complex system design, code architecture, and high-level refactoring.
ChatGPT	Rapid Prototyping & Syntax	Regex generation, bash command quick checks, and one-liner snippets.
Gemini	Large Context & Multimodal	Ingesting massive log files, long spec PDFs, and visual documentation.
Antigravity	Agentic Execution & IDE Partner	Direct workspace file edits, background task execution, and verified builds.

2. Trigger-Based Handoff Rules & Human-AI Boundaries

Non-Delegable Human Tasks (Human Ownership)

- **Architectural Approval:** Deciding overall site structure, proof claims, and target audience alignment.
- **Quality Gate Evaluation:** Auditing AI outputs against accuracy, performance metrics, and zero-hallucination standards.
- **Final Ship Decision:** Pushing commits, submitting portal assignments, and approving production builds.

Delegable AI Tasks (AI-Native Workflow)

- Boilerplate generation and semantic HTML/CSS styling.
- Writing unit test suites and executing local shell verification commands.
- Generating Mermaid sitemap flowcharts and architectural diagrams.
- Automated code linting, bug tracing, and log analysis.

3. Visual Handoff Architecture Flow

■ Task Origin (Feature, Refactor, Bug, Research)

■■■ Docs / API Search

■■■ ■ Perplexity AI (Live Documentation Search)

■■■ Engineering Work

■■■ ■ Task Scale & Depth Routing:

■■■ High Context / Multimodal

■■■ ■ Gemini

■■■ Deep System Architecture

■■■ ■ Claude

■■■ Syntax / Quick Snippets

■■■ ■ ChatGPT

■■■ Agentic IDE Execution

■■■ ■ Antigravity (Central Hub)

■ Antigravity Execution ■■■ ■ Human Quality Bar Audit ■■■ ■ Ship Proof Artifact

4. Pressure-Test Summary & Refinements

Pressure-Test Finding: Unregulated tool switching creates mental fatigue and fragmented context. **Refinement:** Antigravity is designated as the Central Hub, with all external LLM outputs funneling into Antigravity for execution and empirical verification.

5. Submission Pass Criteria Checklist

- [X] **Toolstack Defined:** Detailed roles for Claude, ChatGPT, Gemini, Perplexity, and Antigravity.
- [X] **Clear Boundaries:** Formulated non-delegable human tasks vs delegable AI tasks.
- [X] **Visual Diagram:** Created Mermaid handoff flowchart (handoff_diagram.mermaid).
- [X] **Pressure-Tested:** Evaluated workflow with AI Tutor (pressure_test_log.md).